Comparative Analysis of Traditional Machine Learning and Deep Learning Models for Real-Time Fraud Detection

Overview

This report evaluates traditional machine learning (ML) models (such as Gradient Boosted Trees and Random Forests) versus deep learning (DL) models (such as Autoencoders and Neural Networks) for high-throughput, real-time financial fraud detection systems processing millions of transactions per day.

Comparison

Criterion	Approach A	Approach B
Detection Accuracy	Traditional ML models (e.g., XGBoost, LightGBM) excel on tabular tabular data with engineered features, often yielding high precision and recall on structured financial feeds.	Deep learning models (e.g., Autoencoders) capture intricate non-linear relationships and high-dimensional representations, outperforming linear or shallow methods on complex anomaly patterns with minimal mislabeling.
Computational Cost	Traditional ML models require moderate training resources and lightweight deployment infrastructure, making scaling to millions of daily transactions cost-effective.	Deep learning models demand significant GPU/TPU resources for training, larger memory footprints, and intensive cluster setups to maintain throughput.
Inference Latency	Traditional ML models provide ultra-low inference latencies (often under 5 milliseconds), fitting easily within strict real-time authorization windows.	Deep learning architectures can introduce higher latency overhead due to multi-layer matrix operations, requiring hardware acceleration (e.g., TensorRT, ONNX runtimes) to meet sub-10ms SLAs.

Ability to Detect New Fraud Patterns	Traditional ML models are susceptible to concept drift and require manual feature engineering updates to catch novel scam variations.	Unsupervised deep learning models like Autoencoders can flag out-of-distribution transactions as anomalies, making them inherently more adaptive to novel, zero-day fraud tactics.

Strengths & Weaknesses

Approach A

+ Extremely fast inference times suitable for strict millisecond service-level agreements (SLAs)
+ Lower computational overhead and resource costs for training and serving
+ High interpretability and ease of feature importance extraction (e.g., SHAP values)
+ Robust performance on structured, tabular transaction datasets
– Requires constant manual feature engineering to adapt to evolving fraud vectors
– Vulnerable to rapid concept drift without frequent model retraining
– Struggles to capture complex sequential and spatial-temporal dependencies without specialized aggregations

Approach B

+ Exceptional capacity to uncover hidden patterns and high-dimensional correlations
+ Unsupervised architectures (like Autoencoders) handle novel and zero-day fraud patterns effectively
+ Seamlessly handles unstructured or semi-structured transaction metadata (e.g., device logs, text)
– High inference latency unless optimized via dedicated hardware accelerators
– Substantial computational and infrastructure cost for high-throughput streaming workloads
– Black-box nature complicates regulatory compliance and auditability requirements

Recommendation

For a financial institution processing millions of transactions daily under tight real-time constraints, a hybrid architecture is recommended. Employing an optimized traditional ML model (such as gradient-boosted trees) as a first-tier gatekeeper ensures sub-millisecond inference and low operational costs. Simultaneously, a parallel unsupervised deep-learning anomaly detector (such as an Autoencoder) can monitor for novel zero-day patterns and complex behavioral anomalies. Note that hardware limitations or lack of specialized GPU/TPU accelerators

> in legacy core banking stacks may require software-level graph optimization (e.g., ONNX) before deploying deep models.

Sources

Credit Card Fraud Detection in the Nigerian Financial Sector: A Comparison of Unsupervised TensorFlow-Based Anomaly Detection Techniques, Autoencoders and PCA Algorithm

Demonstrates that deep learning autoencoders offer superior performance in analyzing complex transactional datasets and reducing mislabeling compared to linear dimensionality reduction.

http://arxiv.org/abs/2407.08758v1

Changing Data Sources in the Age of Machine Learning for Official Statistics

Outlines critical risks associated with concept drift and shifting data distributions in high-throughput machine learning monitoring systems.

http://arxiv.org/abs/2306.04338v1